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The Editors
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Dear Editors,

We submit **“Same Prompt, Different Answer: Hidden Non-Determinism in LLM APIs Undermines Scientific Reproducibility”** for consideration as an Article in *Nature Communications*.

Researchers across medicine, climate science, engineering, and social sciences increasingly rely on large language model APIs to extract data, synthesise evidence, and generate hypotheses. Every major provider documents temperature-zero inference as “deterministic.” We show this guarantee is false—and that the failure is invisible, creating a hidden reproducibility threat that cuts across disciplines.

Our 4,104 controlled experiments across eight models and five providers reveal that API-served models reproduce their own outputs only 22% of the time under documented deterministic settings, while locally deployed models achieve 96%. The variation is undetectable by inspection: outputs remain semantically equivalent (BERTScore F1 > 0.97) while differing textually. In multi-turn and retrieval-augmented workflows—now standard in automated evidence synthesis and clinical decision support—reproducibility drops to near zero. A quasi-isolation experiment attributes the cause to production infrastructure complexity (tensor parallelism, dynamic batching), not cloud deployment itself, demonstrating that the problem is solvable.

This work is broadly relevant for three reasons. First, every discipline that uses LLM-generated outputs in published research is affected: thousands of studies implicitly assume deterministic inference without empirical verification. Second, the EU AI Act and NIST AI Risk Management Framework require traceability for high-risk AI systems—our findings show that current API behaviour fails this standard by default. Third, the rapid adoption of agentic AI pipelines, where LLMs are chained across multiple steps, means non-determinism compounds multiplicatively, making provenance tracking essential infrastructure rather than a niche concern.

The real-world impact of this hidden variation is already documented. In a companion study currently under review at *Research Synthesis Methods* (Rover & Tadano, 2026), we apply six LLMs to 500 PubMed abstracts on fine particulate matter and respiratory health. A meta-analytic propagation analysis shows that up to 23 study-level epidemiological findings—each a distinct association between air pollution and health outcomes—appear or disappear from the evidence base depending solely on which run of the LLM is used. This demonstrates that the non-determinism quantified in the present manuscript has direct consequences for scientific conclusions in applied health research.

We provide a practical solution: a lightweight provenance protocol grounded in W3C PROV that adds less than 1% overhead and makes hidden variation detectable. All 4,104 experimental records and the protocol implementation are available at <https://github.com/Roverlucas/genai-reproducibility-protocol> under open licenses (MIT for code, CC-BY 4.0 for data).

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the submission. We declare no competing interests.

Yours sincerely,

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